

The Evolution of Tennis Dominance

Process Book - Milestone 3 – Team nadal-prime

Project of Data Visualization (COM-480)

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Live website: <https://com-480-data-visualization.github.io/nadal-prime/>

1. From an idea to a question

Our starting point was the data, not the question. We came across Jeff Sackmann's ATP repositories, an enormous, well-structured record of professional tennis going back to 1968 and wanted to build the project around them. The question came afterward, out of something that hangs over every sport: who is the greatest of all time? It is a question that is always present and never really settled, and the interesting version of it is comparative across time. Would today's best player dominate in the past, or would the past legends survive in today's game?

We quickly realized the straight version of that question is badly posed. A player from the wooden-racket era and a player from today are not doing the same thing, even if both are holding a racket on the same court, so comparing them head-to-head means little. Instead of trying to build yet another all-time ranking, we reframed the project around a different question: how does the profile of the dominant ATP player change across eras? Rather than producing a single all-time ranking, we wanted to show which features best characterize elite performance in each period, and to connect those changes back to what was happening in the sport: equipment, strings, training, and pace of play.

The working hypothesis we carried from Milestone 1 is that structural changes in tennis shift the set of features that best reflect elite performance, and so that dominance is not static but moves with the game. The intended audience stayed consistent across milestones: casual tennis followers and sports-analytics readers first, but also coaches, equipment engineers, and tennis regulators, who can each see how their decisions ripple into how the game is actually played.

2. The data and how we shaped it

2.1 Sources

We built on two of Jeff Sackmann's public repositories. The *tennis_atp* repository gives match-level files from 1968 to today with structured information on players (ID, age, handedness, height, nationality), competition context (tournament level, surface), and performance indicators such as aces, double faults, serve points, and break points. The *Match Charting Project* adds shot-level and tactical metadata for a large set of manually charted matches: shot type and direction, serve direction and outcome, return depth, and net-point patterns. That second source is what makes the project possible, because our visualizations need point-by-point detail that the standard ATP files, which are match-level only, simply do not contain.

The two sources play complementary roles: *tennis_atp* provides robust long-term coverage and comparability across eras, while the Match Charting Project enables the deeper tactical reading of player profiles. We knew about two limits from the start. Coverage is uneven over time, and the charting is selective, with far fewer charted matches than the full ATP corpus and inconsistent representation across periods and players.

2.2 Preprocessing

Our pipeline restricts the scope to ATP men's singles. The ATP match files from 1968 onward are concatenated into a single table, player metadata and rankings are parsed and merged, and the Match Charting Project stat files are reduced to per-match, per-player totals. The cleaning produced a small set of tidy CSVs that the website reads directly: service, returns, shots, an overview table, and net points from the charting data, plus an all-matches file and a rankings file from the ATP data.

At this stage the charting statistics were grouped by decade. A helper extracts the year from each match identifier and buckets it into ten-year periods (1960-1969, 1970-1979, and so on), which let us look at long-term trends without committing to any particular set of historical boundaries. Those boundaries came later: the five named eras are the backbone of the final site, and we settled them during Milestone 2.

- Open Era Foundations (1968–1979): the wooden racket, serve-and-volley game.
- Graphite Era (1980–1994): graphite frames add power and make the serve a more direct weapon.

- Transition Era (1995–2007): polyester strings and topspin push the game toward the baseline.
- Big 3 Era (2008–2019): Nadal, Federer and Djokovic, more than ten years of reign.
- Modern Era (2020–present): The new generation, where physical and mental strengths are dominant.

The five eras were an editorial choice made for the storytelling: naming the periods makes the shifts in play easier to present than abstract decade buckets would. The earlier boundaries track equipment and technique, the move from wooden to graphite frames around 1980 and the spread of polyester strings and heavy topspin through the mid-1990s. The later boundaries track who was winning, the collective peak of Federer, Nadal, and Djokovic from 2008 and the changing of the guard toward a younger generation from 2020.

The feature groups that structure the final site were already present at this stage as separate tables, just organized differently. Service, shots, overview, and net points carried through into the final three lenses (Service, Rally, and Finishes), and the spine of the story was already visible in the early statistics. Net points fall steadily from the 1960s while the net success rate rises (Table 1), which says players moved to the baseline and now come forward only when it pays off. That single trend is what the Rally and Finishes lenses go on to develop in detail.

	period	mean_net_points	net_success_rate
0	1960-1969	103.600000	0.597490
1	1970-1979	71.940000	0.624131
2	1980-1989	69.126697	0.632879
3	1990-1999	53.000000	0.654147
4	2000-2009	28.445479	0.664496
5	2010-2019	19.090045	0.682927
6	2020-2029	19.931778	0.687203

Table 1: Mean net points per match and net-point success rate by decade, from the Milestone 1 exploratory analysis. Net points fall sharply across the decades while the success rate climbs, an early signal of the shift from serve-and-volley toward a baseline game that the final visualization builds on.

3. Designing the story

From Milestone 2 onward we committed to a scroll-based visual story rather than a dashboard. Our central claim is that dominance is not static, and a guided scroll lets us pair each chart with the historical reason behind it.

3.1 Structure of the site

The site has three parts:

- Landing: a full-screen title page shown before any data appears.
- Defining the Eras: five era cards, each with a short historical explanation (the wooden age, the graphite revolution, the baseline shift, the era of excellence, the next generation) and an "average player profile" showing mean age, height, and match duration for that era. Hovering an era card reveals its Top-5 ranking history as a line chart.
- Playstyle Evolution: the analytical core. A point is split into three phases, Service, Rally, and Finishes, and a feature selector lets the reader switch between them. Each phase has a scatter plot relating a style metric to ATP points, plus a second view suited to that phase, all driven by a sticky scroll timeline of the five eras. Once the user reaches the bottom of a feature, clicking another feature scrolls them back to the top of the section. The scroll only happens the first time each feature is opened, so the user can then switch between features to compare an era across lenses without scrolling up again.

3.2 The three lenses

Each lens answers “what does dominance look like through this phase of the point?”

- Service: scatter of the share of points won in three shots or fewer against ATP points, paired with a service heat map showing where serves land and where they miss (with In/Out toggles).
- Rally: scatter of volley-shot percentage against ATP points, paired with a horizontal bar chart of the full shot-type distribution (groundstrokes, slices, volleys, lobs, overheads, drop shots, swing volleys).
- Finishes: scatter of mean winners per match against ATP points, paired with a pie chart of how points actually end (aces, forehand/backhand winners, net winners, passing winners, and forced opponent errors).

Each dot in the scatter plot is one player within one era. As the reader scrolls into an era, that era's players are emphasized while the others fade, and the explanatory text and chart titles update at the same time. We had wanted a scroll-based interaction to move through time since Milestone 2, and keeping the other eras on screen rather than hiding them was a deliberate choice so that eras stay easy to compare.

3.3 From sketches to the final design

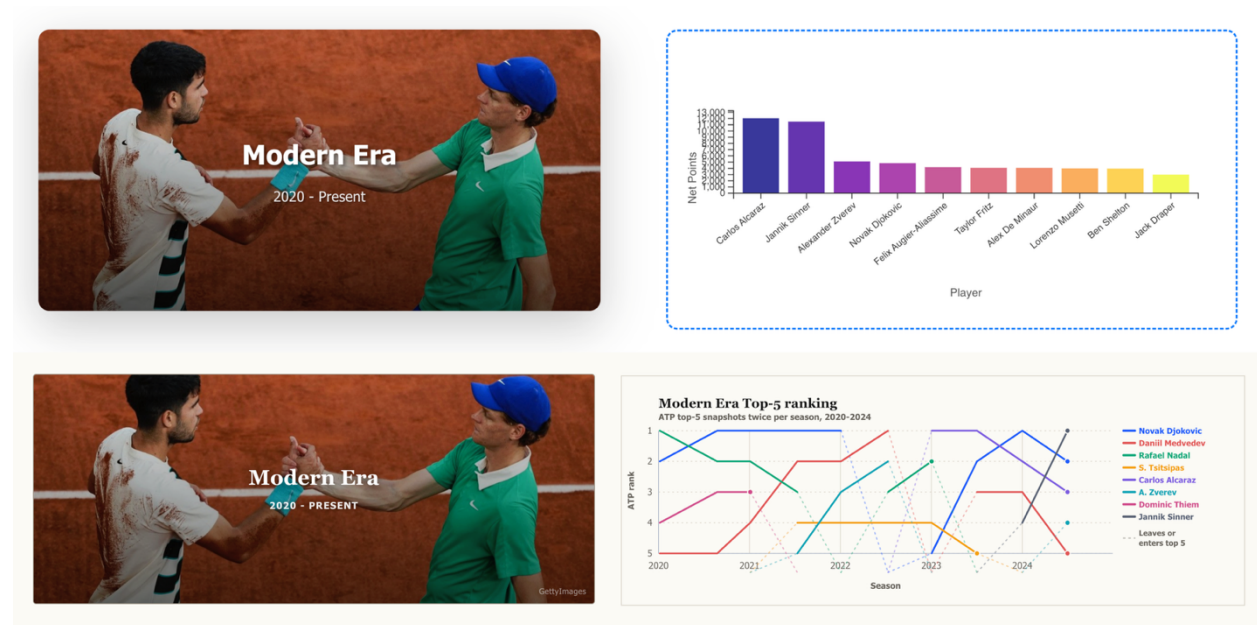


Figure 1: The era ranking view. The Milestone 2 bar chart of top players by aggregated points (top) became a Top-5 ranking history line chart also revealed on hover over each era card (bottom).

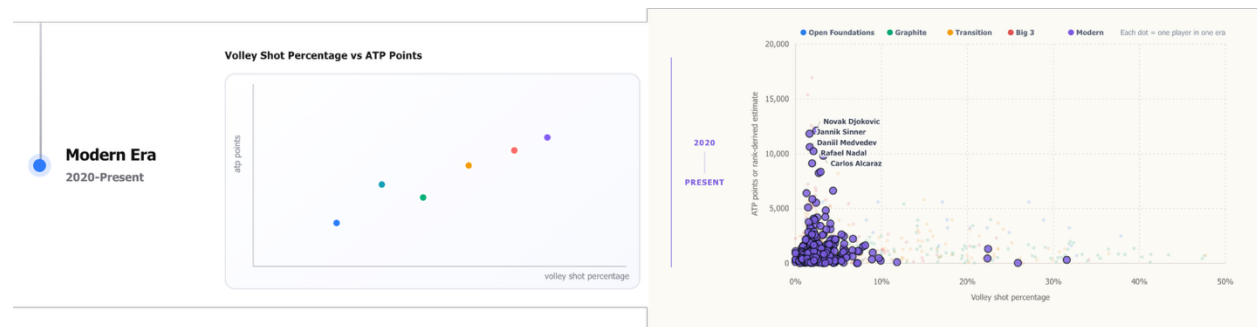


Figure 2: The volley scatter, Milestone 2 prototype (left) versus the final site (right).

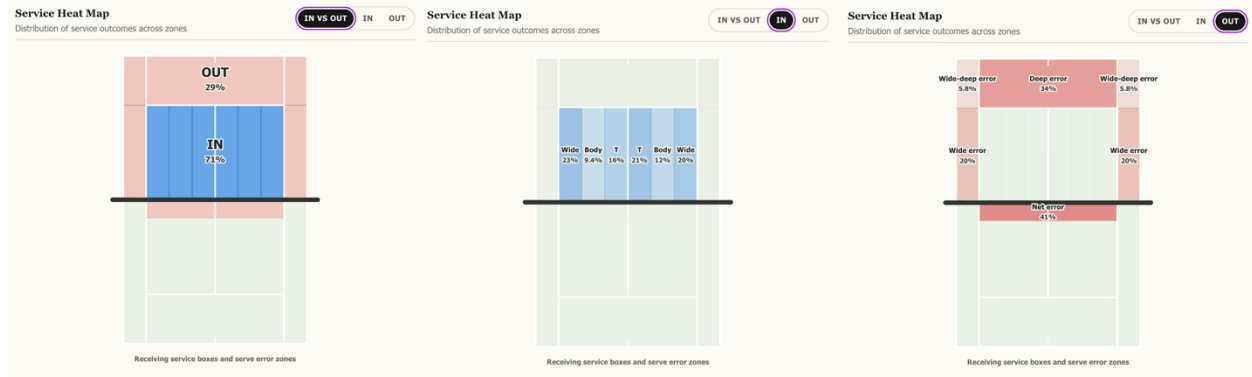


Figure 3: The service heat map, added after Milestone 2. Serves are mapped onto a court by landing zone, with separate views for serves that land in and serves that miss, giving a direct read on the service phase.

Some of what we sketched in Milestone 2 changed once we started building:

- Sankey diagrams were dropped. In Milestone 2, we planned a shot-type-to-point-outcome Sankey (Figure 3 of Milestone 2). We replaced it with the Finishes pie chart and the service heat map, which carried the "how points end" and "where serves go" ideas more directly and were easier to read visually.
- The spider and parallel-coordinates charts (Figures 4 and 5 of Milestone 2) were also dropped. They were listed in Milestone 2 as extra ideas rather than part of the core, and once the three lenses took shape they turned out to overlap with what the scatter plots and distribution views already showed, while being harder to fit cleanly into the scroll layout.
- The bar chart of top players by aggregated points (Figure 1) became the per-era Top-5 ranking line chart shown on hover.
- The service heat map (Figure 3) was not in the Milestone 2 plan at all. It came later, and it fit the project well thematically: showing where serves land and where they miss on an actual court is a natural way to read the service phase of a point.

4. Technical implementation

The site is a static page: a single index.html, one stylesheet, and one script.js using D3 v7 loaded from a CDN. Keeping it static means it deploys straight to GitHub Pages with no build step. All data lives in CSV files under *datasets/clean_datasets/*, fetched at runtime.

4.1 How the charts are built

The charts are written directly in D3 with SVG. This gave us the control we needed over the scroll-linked transitions, where switching era has to update several charts at once without redrawing them from scratch. The main parts of script.js are:

- A data-preparation step that reads the CSVs once and aggregates them into per-player, per-era profiles for each lens (*prepareServiceData*, *prepareRallyData*, *prepareFinishesData*).
- A player-strength lookup, built from the all-matches file, that stores each player's peak ranking points and best rank per era. This is also where the rank-based point estimate is used when a ranking-points value is missing (*rankToPointEstimate*).
- A scroll controller based on *IntersectionObserver*. As each era enters the viewport it becomes the active era, which updates the scatter emphasis, the second chart, and the text panel together.
- A set of reusable chart builders: one scatter builder shared by all three lenses, plus a bar chart, a pie chart, and a court heat map drawn from real court proportions. Each has its own update function, so changing era animates the existing chart instead of rebuilding it.
- The Top-5 ranking charts on the era cards, which stream and parse the rankings CSV, sample two snapshots per season, and draw each player's path, with separate lines marking when a player enters or leaves the top 5.

4.2 Challenges

A few problems took some work to get right.

- Sparse and uneven data. Charting coverage is thin in the early eras, so naive averages were noisy. The rank-based point estimate and the per-era aggregation were our way of keeping the early eras (especially the Wooden era which doesn't have ATP ranking points) present without overstating how precise they are.
- Keeping text and charts in sync while scrolling. Getting the right era to activate at the right scroll position meant finessing the *IntersectionObserver* margins, and we had to avoid flicker on the sticky header.
- Label collisions in the scatter. Player labels overlapped badly, so we added a small placement routine that spreads them vertically and picks which side of a point to anchor each label.

- Performance when switching eras. We replaced per-point SVG attribute updates with CSS class toggles, which is smoother. We choose to keep all players points visible, accepting a small performance cost, so that every player stays comparable across eras.
- Responsive layout. We had to make sure the layout would not break for lower resolutions screens. We had to get all the styles right to make sure the visualizations and most importantly the scrolling behavior would not break for smaller screen. We also added a warning that show once the vertical height was too small for the layout to work. Making the website work on mobile would require major refactors of the code and our original scrolling idea.

5. Peer assessment

Valentin Belardi: Rally and finishes feature family, scatter plots, rally shot distribution, point-ending visualization, explanatory text, README.

Luca Deandrea: Service feature family, service scatter plot, tennis-court heat map, service outcome selector, scroll interactions, smooth visual updates, screencast.

Martin Sanchez Lopez: Defining the Eras section, historical framing, average player profile, ranking time-series visualizations, era narrative structure.